Fitted survival models and the molecular international prognostic scoring system in myeloid neoplasms: a centre-grouped validation study with an unseen external centre

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Abstract

Background: Machine-learning survival models are increasingly proposed for prognosis in myeloid neoplasms, but they are rarely compared with the current molecular standard, the Molecular International Prognostic Scoring System (IPSS-M), under a validation design that mirrors use at a new hospital.

Methods: I analysed 3,173 patients from 23 centres (1,600 deaths) released in a public prediction challenge, and an external centre of 1,193 patients whose outcomes were withheld. Predictors were blood counts, a rule-based reading of free-text karyotypes, and somatic mutations restricted to the 64 genes sequenced at both the training and the external centres. The published revised score (IPSS-R) and the IPSS-M, applied without refitting, were compared with Cox models, an elastic-net Cox model, a random survival forest, gradient-boosted Cox trees and a rank-averaged ensemble. Discrimination was measured by the censoring-weighted concordance index truncated at 7 years, over 15 centre-grouped cross-validation folds with nested tuning and corrected confidence intervals, and on the external centre.

Results: Without fitting, the IPSS-M reached a concordance of 0.720 (95% confidence interval 0.693 to 0.748), against 0.669 for the IPSS-R. The best fitted model, an ensemble of the elastic-net and gradient-boosted Cox models chosen before external scoring, reached 0.725 (0.700 to 0.750) and beat the elastic-net alone on all 15 folds, but its advantage over the IPSS-M was small and uncertain (+0.005, −0.005 to +0.014; p = 0.29). A recalibrated IPSS-M was slightly better calibrated (integrated Brier score 0.160 against 0.163). On the external centre, the ensemble scored 0.755 and the IPSS-M 0.726, a gap unexplained by case mix or missing data. Mutation burden on the shared gene panel was the most important predictor group.

Conclusions: Fitted models matched the IPSS-M internally and exceeded it at one external centre. New prognostic models for myeloid neoplasms should be benchmarked against the IPSS-M with centre-level validation before improvement is claimed.

Introduction

Myeloid neoplasms, mainly the myelodysplastic syndromes (MDS) and acute myeloid leukemia, range from slowly progressing disease to rapidly fatal illness. Estimating a patient's prognosis at diagnosis guides treatment: patients at low risk may receive supportive care, while patients at high risk may be considered for intensive therapy such as allogeneic stem cell transplantation. For MDS, risk has long been estimated with the

revised International Prognostic Scoring System (IPSS-R), which combines the karyotype, bone-marrow blasts and three blood counts [1, 2]. Its molecular successor, the IPSS-M, adds somatic mutations in 31 genes and was developed and validated on several thousand patients [3].

At the same time, machine-learning survival models such as random survival forests [4] and gradient-boosted trees [5, 6] are often proposed as improvements over regression. Two features of the evidence call for caution. First, systematic reviews of clinical prediction models have found no average performance benefit of machine learning over regression when both are developed carefully [7]. Second, new models are frequently compared with weak baselines and validated on random splits of a single dataset, which mixes patients from the same hospital into the training and validation sets; such internal validation overstates performance at a new hospital, where case mix and measurement practice differ [8, 9].

A public data challenge organised by Qube Research and Technologies (QRT) and Institut Gustave Roussy offers a setting to test both points [10]. It provides clinical, cytogenetic and molecular data on patients from 23 centres, and scores predictions on a further centre whose outcomes are never released. A similar crowdsourced effort, the DREAM 9 acute myeloid leukemia challenge, also scored survival models by a concordance index [11].

The aim of this study was to ask whether fitted survival models, from penalised Cox regression to boosted trees and their ensemble, improve on the published IPSS-M when validated on hospitals they were not trained on. The study is reported following the TRIPOD+AI guideline [12] (S1 Table).

Materials and methods

Data source and participants

The data are those of challenge 162 on the Challenge Data platform (École normale supérieure, Paris) [10]. The training set contains 3,323 patients from 23 clinical centres, with overall survival in years and a death indicator. The test set contains 1,193 patients, all from one further centre (coded KYW) that contributes no training patients; their outcomes are withheld, and predictions for them are scored by the platform. One hundred and fifty training patients have no recorded outcome and were excluded, leaving 3,173 patients and 1,600 deaths. The median blast count of 3% indicates that most patients have MDS. Age, sex, disease subtype and treatment are not provided.

Ethics statement

This is a secondary analysis of de-identified data distributed by the challenge organisers to registered participants. No new data were collected, and the author had no access to identifying information. [TO CONFIRM WITH THE ORGANISERS: the name of the ethics committee that approved the original data collection, whether patient consent was written, and written permission to publish results derived from the challenge data.]

Outcome

The outcome was overall survival, the time from diagnosis to death from any cause, with patients alive at last follow-up treated as right-censored. Fifty-three patients had an observed time of zero; all were censored and left the risk set immediately.

Predictors

Ninety-one candidate predictors were built from each patient's own data; full definitions are given in S1 Appendix.

Clinical variables. Bone-marrow blasts (%), haemoglobin (g/dL), and the logarithm of one plus white cells, neutrophils, monocytes and platelets (G/L), with a 0/1 flag for each missing value. Missing values were replaced by the median of the training part of each fold, inside the model pipeline.

Karyotype. Karyotypes are free text in International System for Human Cytogenomic Nomenclature notation, written differently by different centres (lower case in training, upper case and the word "Normal" at the test centre). A rule-based parser normalised the text and extracted the number of abnormalities, 14 recurrent lesions (including monosomy 7, deletion 5q, trisomy 8 and 17p loss), complex and monosomal status, and the IPSS-R cytogenetic category [2]. It was checked on 25 hand-written test strings.

Mutations. The training file lists 124 mutated genes and the test file 83, of which only 64 appear in both; several genes frequent in training never appear at the test centre, which suggests a smaller sequencing panel there. To give mutation counts the same meaning at every centre, all molecular features used only the 64 shared genes. Features were the numbers of mutations and of mutated genes, the maximum variant allele fraction (VAF), the number of truncating mutations, TP53 count, maximum VAF and multi-hit status, FLT3 internal tandem duplication, MLL partial tandem duplication, six pathway counts, and indicators for the 45 shared genes mutated in at least 1% of training patients.

Benchmark scores

Both benchmark scores were computed from their published definitions, with no parameter estimated from these data.

IPSS-R. The sum of points for cytogenetic category, blasts, haemoglobin, platelets and neutrophils [1]; a missing component received the median points of the training patients.

IPSS-M. The IPSS-M is a weighted sum of centred variables,

$$\text{IPSS-M} = \frac{1}{\ln 2} \sum_{k} \beta_k \left(x_k - \mu_k \right), \tag{1}$$

where β_k is the published coefficient of variable k and μ_k its published reference mean [3]. The coefficients and means were taken from the authors' open-source calculator [13]. A missing input was set to its reference mean, so that it contributes nothing to the sum. The implementation agreed exactly with the official calculator on 300 training patients with complete data. Copy-neutral loss of heterozygosity is not available in these data, so a single TP53 mutation counted as multi-hit only when its VAF exceeded 0.55 or a 17p loss was present.

Models

All fitted models used the same predictors and were fitted with the same procedure in every fold.

Cox models. The Cox proportional hazards model [14] writes the hazard of a patient with predictors x as

$$h(t \mid x) = h_0(t) \exp\left(\beta^\top x\right), \tag{2}$$

and the linear predictor $\beta^\top x$ is the risk score. Three versions were fitted: a four-variable model (blasts, haemoglobin, platelets and number of mutations) reconstructed from the challenge description; an IPSS-M-style model with 24 variables chosen from the literature (the five blood values, the IPSS-R cytogenetic category, a missing-karyotype flag, the 16 IPSS-M main-effect genes and the capped count of IPSS-M residual genes), with a small ridge penalty of 0.1; and an elastic-net Cox model [15,16] on 84 predictors (all except monocytes and the missing flags), which maximises

$$\ln L(\beta) - \alpha \left(\rho \sum_k |\beta_k| + \frac{1-\rho}{2} \sum_k \beta_k^2 \right), \tag{3}$$

where L is the Cox partial likelihood. The penalty strength α (30 values) and mix $\rho \in \{0.1, 0.5, 1\}$ were tuned by nested cross-validation.

Tree models. A random survival forest [4] with 300 trees, at least 15 patients per leaf and nine predictors tried per split; and gradient-boosted trees with a Cox loss (XGBoost [6]; learning rate 0.03, 80% of patients and 50% of predictors sampled per tree, minimum child weight 10). The boosted model's depth (2, 3 or 4) and number of trees (50 to 600) were tuned by nested cross-validation, and five models with different random seeds were averaged. Exact split finding on a single thread was used, so that results are reproducible on one machine.

Ensemble. Because the metric depends only on the order of the risk scores, each model's scores were converted to ranks between 0 and 1 and averaged. The ensemble of the elastic-net and boosted Cox models was specified as the final model before any of its external scores were known.

Validation

Centre-grouped cross-validation. All patients of a centre were placed in the same fold, so that every validation fold contained only centres unseen in training. The 23 centres were dealt into five folds of similar size in a random order, and this was repeated three times with different orders, giving 15 validation folds. All tuning was nested: the penalty and tree settings were chosen by an inner four-fold split of the training part, also grouped by centre, and median imputation and standardisation were learned on the training part only. Every model was evaluated on the same 15 folds, so that models could be compared fold by fold.

External validation. Models refitted on all 3,173 training patients were applied to the test centre and scored by the challenge platform on its public subset of patients. Submissions are limited to two per day. Before each of the last submissions, the expected score was written down: the public score of the elastic-net plus the model's cross-validated paired difference with the elastic-net.

Performance measures

Discrimination was measured by the inverse-probability-of-censoring-weighted concordance index [17], the challenge metric. For patients i with observed time T_i,

death indicator Δ_i and risk score η_i,

$$C_\tau = \frac{\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau) \, \mathbf{1}(\eta_i > \eta_j)}{\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)}, \tag{4}$$

where $\hat{G}$ is the Kaplan–Meier estimate [18] of the probability of remaining uncensored, estimated on the training part, and $\tau = 7$ years. A pair started by a death at a time when few patients remain under follow-up is rare to observe and so receives a larger weight. Harrell's concordance index [19] was also recorded. The cumulative/dynamic area under the receiver operating characteristic curve at 1, 3 and 5 years [20] was computed on all 15 folds, with validation follow-up censored at 6.5 years so that the censoring weights stay defined. Calibration was assessed on the five folds of the first repetition by the Brier score at 3 years and integrated from 0.25 to 6 years [21], and by comparing mean predicted with Kaplan–Meier observed 3-year survival in ten groups of predicted risk [22]. For this comparison the IPSS-M was recalibrated by a one-variable Cox model on its score, fitted on each training part, so that its published weights were kept and only the link from score to probability was estimated; the elastic-net used its Breslow baseline survival.

Statistical analysis

The 15 cross-validation scores are not independent, because the three repetitions reuse the same patients and training sets overlap. Confidence intervals and two-sided p-values for mean scores and paired differences therefore used the corrected resampled t-test of Nadeau and Bengio [23], which inflates the variance of the mean to

$$\widehat{\mathrm{Var}} = \left(\frac{1}{J} + \frac{n_{\mathrm{valid}}}{n_{\mathrm{train}}} \right) s^2 = \left(\frac{1}{15} + 0.257 \right) s^2 = 0.323 \, s^2, \tag{5}$$

where $J = 15$ is the number of folds, s the standard deviation of the fold scores, and 0.257 the average ratio of validation to training patients, with a t distribution on 14 degrees of freedom. The proportional hazards assumption of the IPSS-M-style Cox model was checked with scaled Schoenfeld residuals [24]. Permutation importance [25] was measured on held-out folds by shuffling groups of related predictors together and recording the fall in concordance.

To interpret the difference between the ensemble and the IPSS-M at the external centre, three explanations were tested on the training data. For case mix, each validation fold was resampled 20 times without replacement to the external centre's distribution of IPSS-R risk groups. For missing data, blood counts and blast percentages were removed for an extra 7% of each validation fold, matching the higher missingness at the external centre, and the fall in concordance of the IPSS-M and the elastic-net was compared. For between-centre variation, the difference was computed within each training centre with at least 100 patients, using out-of-fold predictions.

Software

Analyses used Python 3.13 with pandas [VERSION], NumPy [VERSION], scikit-learn [VERSION] [26], scikit-survival [VERSION] [27], lifelines [VERSION] [28] and XGBoost 3.4.1 [6]. All code is available at https://github.com/bernardfrimpong040-pixel/aml-survival-prediction [ARCHIVE ON ZENODO AND ADD THE DOI BEFORE SUBMISSION].

Use of artificial intelligence tools

An AI assistant (Claude, Anthropic) was used to help write and review analysis code and to edit the manuscript text. The author designed the study, ran every analysis, checked all results and takes full responsibility for the content.

Results

Cohort

Table 1 describes the two cohorts. The median overall survival in training was 3.15 years. The test centre was clinically sicker than the training centres: it had more blasts, lower haemoglobin and platelets, almost three times the share of complex karyotypes, and nearly three times the share of TP53 multi-hit patients. One third of its patients were IPSS-R very high risk, against 11% in training. Monocytes were missing for 74% of test patients.

Table 1. Characteristics of the training cohort and the external test centre.

Characteristic	Training (23 centres)	Test (one centre)
Patients	3,173	1,193
Sex inferred from karyotype: female / male / unknown, %	33.3 / 52.7 / 14.0	17.1 / 45.8 / 37.1
Deaths, n (%)	1,600 (50.4)	withheld
Median overall survival, years	3.15	withheld
Median follow-up (reverse Kaplan–Meier), years	3.84	withheld
Bone-marrow blasts, %	3.0 (1.0–8.0)	6.0 (2.0–17.0)
Haemoglobin, g/dL	9.7 (8.5–11.1)	8.4 (7.4–10.1)
Platelets, G/L	123 (65–230)	74 (37–143)
White blood cells, G/L	4.1 (2.7–6.7)	2.9 (1.9–5.3)
Neutrophils, G/L	2.00 (1.00–3.70)	1.30 (0.54–2.64)
Karyotype missing or unusable, n (%)	350 (11.0)	120 (10.1)
Normal karyotype, n (%)	1,643 (51.8)	410 (34.4)
Complex karyotype, n (%)	307 (9.7)	336 (28.2)
IPSS-R cytogenetics poor or very poor, n (%)	408 (12.9)	380 (31.9)
Mutated genes per patient (shared panel)	2 (1–4)	2 (1–3)
TP53 multi-hit, n (%)	210 (6.6)	213 (17.9)
SF3B1 mutated, n (%)	727 (22.9)	92 (7.7)
IPSS-R very high risk, n (%)	363 (11.4)	395 (33.1)
IPSS-M very high risk, n (%)	666 (21.0)	494 (41.4)

Values are median (first–third quartile) unless stated. Sex is not recorded in the data and was inferred from the sex chromosomes in the karyotype. Outcomes of the test centre are withheld by the challenge.

Both benchmark scores separated survival in the expected order. The six IPSS-M categories gave a log-rank chi-squared statistic of 974.2 on 5 degrees of freedom, against 737.4 on 4 for the five IPSS-R groups, with median survival falling from 9.65 years in the IPSS-M very low category to 1.03 years in the very high category (Fig 1). At the test centre, 41% of patients were IPSS-M very high risk, against 21% in training.

Fig 1. Survival by IPSS-M category and case mix of the two cohorts. Left: Kaplan–Meier overall survival by IPSS-M category in the 3,173 training patients. Right: share of patients in each IPSS-M category in training and at the external test centre.

Discrimination in centre-grouped cross-validation

Table 2 gives the concordance of every model over the 15 centre-grouped folds. The IPSS-R reached 0.669, and its spread across folds (standard deviation 0.029, range 0.624 to 0.729), which involves no fitting, shows how much the difficulty of the task varies between centres. The IPSS-M, also without fitting, reached 0.720. Of the fitted models, the four-variable Cox model reached 0.688 and the IPSS-M-style Cox model 0.716. The elastic-net reached 0.722, the random survival forest 0.719, the boosted Cox model 0.724 and the ensemble 0.725. Adding monocytes and the missing-value flags to the elastic-net changed its concordance by −0.0004 and lost 11 of 15 folds, so these variables were left out.

Table 2. Concordance over 15 centre-grouped cross-validation folds.

Model	Fitted	IPCW C-index (95% CI)	SD over folds	Harrell's C
IPSS-R score	No	0.669 (0.634–0.703)	0.029	0.692
IPSS-M score	No	0.720 (0.693–0.748)	0.023	–
Four-variable Cox	Yes	0.688 (0.653–0.723)	0.029	0.703
IPSS-M-style Cox (24 variables)	Yes	0.716 (0.688–0.743)	0.023	0.736
Elastic-net Cox (84 predictors)	Yes	0.722 (0.696–0.747)	0.021	0.740
Random survival forest	Yes	0.719	0.021	–
Boosted Cox trees (XGBoost)	Yes	0.724 (0.700–0.749)	0.020	–
Ensemble: elastic-net and boosted Cox	Yes	0.725 (0.700–0.750)	0.021	–

IPCW C-index, concordance index weighted by the inverse probability of censoring, truncated at 7 years. Confidence intervals use the corrected resampled t-test (Eq 5). SD, standard deviation. Dashes: not computed.

Because all models were scored on the same folds, paired differences remove most of this between-centre variation (Table 3, Fig 2). The gains from clinical knowledge were large and certain: the IPSS-M beat the IPSS-R by 0.052 (95% CI 0.037 to 0.066), and the 24 literature-chosen variables beat the four-variable model by 0.028 (0.010 to 0.045), winning 14 of 15 folds. The gains from more flexible algorithms were small. The ensemble beat the elastic-net by 0.003 (0.002 to 0.005) on all 15 folds; the boosted model alone beat it by 0.003 (−0.000 to 0.005) on 14 of 15. No fitted model was distinguishable from the IPSS-M: the ensemble was ahead by 0.005 (−0.005 to 0.014; p = 0.29), winning 11 of 15 folds, and the elastic-net by 0.002 (−0.007 to 0.011; p = 0.71).

Table 3. Paired differences in concordance over the same 15 folds.

Comparison	Mean difference (95% CI)	p	Folds won
IPSS-M minus IPSS-R	0.052 (0.037 to 0.066)	<0.001	15/15
IPSS-M-style Cox minus four-variable Cox	0.028 (0.010 to 0.045)	0.004	14/15
Elastic-net minus IPSS-R	0.053 (0.033 to 0.073)	<0.001	15/15
Elastic-net minus IPSS-M	0.002 (−0.007 to 0.011)	0.71	10/15
Boosted Cox minus elastic-net	0.003 (−0.000 to 0.005)	0.07	14/15
Ensemble minus elastic-net	0.003 (0.002 to 0.005)	<0.001	15/15
Ensemble minus IPSS-M	0.005 (−0.005 to 0.014)	0.29	11/15

Positive values favour the first model. Confidence intervals and two-sided p-values from the corrected resampled t-test with 14 degrees of freedom.

Fig 2. Paired differences in concordance with corrected 95% confidence intervals. Each row is the mean, over the same 15 centre-grouped folds, of the difference in the censoring-weighted concordance index between two models.

The time-dependent area under the curve told the same story (S4 Fig). At 1, 3 and 5 years, the fitted models and the IPSS-M were within 0.006 of each other (for the ensemble 0.779, 0.805 and 0.806; for the IPSS-M 0.777, 0.803 and 0.803), and 0.05 to 0.08 above the IPSS-R.

What the models use

In the IPSS-M-style Cox model refitted on all training patients, TP53 multi-hit status had the largest hazard ratio (3.21, 95% CI 2.59 to 3.98), followed by MLL partial tandem duplication (2.05) and FLT3 mutation (1.65), while higher haemoglobin (0.77 per standard deviation) and platelets (0.83) were protective (Fig 3). SF3B1 mutation, strongly protective in a univariable comparison, was neutral after adjustment (1.02, 0.88 to 1.18). Four of 24 variables departed from proportional hazards at $p < 0.05$; the largest departure was NPM1 mutation, harmful in the first months and protective after about four years (S3 Fig).

Fig 3. Hazard ratios of the IPSS-M-style Cox model. Hazard ratios with 95% confidence intervals, refitted on all 3,173 training patients. Blood values are per standard deviation; the cytogenetic category and the residual-gene count are per step; genes are mutated against not mutated.

Permutation importance on held-out centres identified mutation burden (the numbers of mutations and mutated genes, the maximum VAF and the number of truncating mutations, all counted on the shared panel) as the most important predictor group for both the elastic-net and the boosted model, with a fall in concordance of about 0.04 when shuffled, more than twice that of any other group (Fig 4). The boosted model depended less on the TP53 columns than the elastic-net (0.007 against 0.016), consistent with the strong overlap between TP53 multi-hit status and complex karyotype: 88.5% of TP53 multi-hit patients had a complex karyotype, against 4.7% of the others.

Fig 4. Permutation importance on held-out centres. Fall in the censoring-weighted concordance index when the predictors of one group are shuffled together in the validation fold (mean and standard deviation over the five folds of the first repetition), for the elastic-net and the boosted Cox model.

Calibration

Both the recalibrated IPSS-M and the elastic-net reduced the Brier score by about a quarter compared with a Kaplan–Meier curve that ignores all predictors (Table 4). The recalibrated IPSS-M was slightly better calibrated. The elastic-net's predictions were compressed towards the middle: its predicted 3-year survival spanned 0.08 to 0.74 across the ten risk groups, while observed survival spanned 0.08 to 0.85, so patients at low risk did better than predicted (Fig 5). Such compression is the expected effect of penalisation and could be corrected by recalibrating the linear predictor.

Table 4. Brier scores and calibration (five folds of the first repetition).

Model	Integrated Brier score	Brier score at 3 years	Calibration error at 3 years
Kaplan–Meier, no predictors	0.211	0.243	–
IPSS-M, recalibrated	0.160	0.178	0.050
Elastic-net Cox	0.163	0.181	0.065

Lower is better. The integrated Brier score averages the Brier score from 0.25 to 6 years. Calibration error is the mean absolute difference between predicted and Kaplan–Meier observed 3-year survival over ten groups of predicted risk.

Fig 5. Calibration at 3 years. Mean predicted against Kaplan–Meier observed 3-year survival in ten groups of out-of-fold predicted risk, for the recalibrated IPSS-M and the elastic-net Cox model. The diagonal marks perfect calibration.

External validation

Table 5 lists every submission to the test centre. The ensemble, specified as the final model before submission, scored 0.7552, the highest of all submissions and within 0.0023 of its written prediction. The boosted model alone scored 0.7504, below the elastic-net (0.7543) although it had beaten the elastic-net on 14 of 15 folds; a subgroup analysis did not support the explanation that trees rank high-risk patients poorly, since in the IPSS-R very-high-risk subgroup the boosted model ranked better than the elastic-net in cross-validation (0.623 against 0.608, 11 of 15 folds). The IPSS-M scored 0.7255, 0.030 below the ensemble; no prediction was written down before this submission. The fitted models' public scores were 0.022 to 0.033 higher than their cross-validated means (Fig 6), plausibly because the wider spread of risk at the test centre makes patients easier to rank, but the IPSS-M gained only 0.005. Even the IPSS-M-style Cox model, with the same variables as the IPSS-M but weights fitted to the training data, scored higher at the test centre (0.7374), although it was lower in cross-validation (0.716).

The ensemble's lead over the IPSS-M was therefore six times larger at the external centre than in cross-validation (0.030 against 0.005). Neither case mix nor missing data explained this (Table 6). Validation folds resampled to the external centre's risk mix, where 33% of patients are IPSS-R very high risk against 11% in training, gave the same difference as before (0.005). Removing clinical values for an extra 7% of patients lowered the IPSS-M and the elastic-net almost equally (by 0.0026 and 0.0023). Within single training centres, the difference ranged from −0.014 to 0.017 (standard deviation 0.011), and the IPSS-M was ahead in three of eight centres, including the largest (Fig 7). The external centre's difference lay above this range.

Table 5. External validation on the unseen test centre.

Model	Cross-validated C	Written prediction	Public score
Elastic-net Cox	0.7218	–	0.7543
IPSS-M-style Cox	0.7159	–	0.7374
Ensemble (final model)	0.7250	0.7575	0.7552
Boosted Cox trees	0.7244	0.7569	0.7504
IPSS-M score (not fitted)	0.7202	–	0.7255

Public score: censoring-weighted concordance index computed by the challenge platform on the public subset of the test centre. Written prediction: the elastic-net's public score plus the model's cross-validated paired difference with the elastic-net, written down before submission.

Fig 6. Cross-validated against external concordance. Each point is one submitted model. Hollow markers show the predictions written down before submission; dotted lines join them to the public scores.

Table 6. Why is the ensemble further ahead of the IPSS-M at the external centre?

Analysis	Ensemble minus IPSS-M	Detail
Cross-validation, 15 folds	0.005	ensemble ahead on 11/15 folds
Folds resampled to the external case mix	0.005	ensemble ahead on 10/15 folds
Extra 7% of clinical values removed	$+0.0003^{a}$	change in the difference
Single training centres ($\geq$100 patients)	-0.014 to 0.017	SD 0.011, 8 centres
External centre (public score)	0.030	0.7552 minus 0.7255

Censoring-weighted concordance index truncated at 7 years. [a]Elastic-net minus IPSS-M, the change caused by removing the values.

Fig 7. The difference between the ensemble and the IPSS-M varies between centres. Ensemble minus IPSS-M concordance within each training centre with at least 100 patients (out-of-fold predictions; number of patients in brackets) and at the external centre (public score). The dashed line is the cross-validated difference.

Discussion

In 3,173 patients from 23 centres, with validation on held-out centres and on an unseen external centre, carefully built survival models reproduced the discrimination of the published IPSS-M but did not clearly exceed it. The IPSS-M, used with its published weights and no fitting, reached a cross-validated concordance of 0.720; the best fitted model, an ensemble of an elastic-net and a boosted Cox model, reached 0.725, an advantage whose confidence interval includes zero. The recalibrated IPSS-M was also slightly better calibrated. At the external centre, however, the ensemble was 0.030 ahead of the IPSS-M. Two readings fit this result. If the IPSS-M was developed partly on the training centres, its cross-validated scores are optimistic, and a centre outside its development cohort shows its true level; the IPSS-M-style model, refitted to the training data, also overtook it there. Alternatively, the difference simply varies widely between hospitals, as it did across training centres, and this centre lies at the far end. A single external centre cannot separate the two.

Most of the predictive performance came from clinical knowledge rather than from the algorithm. Moving from four generic variables to 24 variables chosen from the IPSS-R and IPSS-M literature added 0.028; every later step, namely penalisation of 84 predictors, boosted trees and the ensemble, added at most 0.006 more. This agrees with systematic evidence that flexible algorithms rarely improve on well-specified regression in clinical prediction [7]. The most important predictors, mutation burden, cytogenetics and TP53, are those already built into the IPSS-M. The modest gain of the boosted model, and the small trees preferred by its tuning, suggest that little structure remains for trees to find once expert-designed variables such as the cytogenetic category and TP53 multi-hit status are present.

Two practical points arose about validation across centres. First, harmonisation mattered: the test centre sequenced a smaller gene panel and recorded karyotypes differently, and mutation counts were the strongest signal only once restricted to genes sequenced at every centre. Without this step a mutation count would have meant different things in training and at the test centre. Second, between-centre variation was

large compared with the differences between models. The unfitted IPSS-R ranged from 0.624 to 0.729 across folds, so comparisons of mean scores would have been uninformative; paired comparisons on the same folds, with confidence intervals corrected for the overlap between folds [23], were needed to separate a consistent 0.003 gain from noise. The external centre also showed that a model winning 14 of 15 internal folds can still lose at a new hospital, which is the protection that averaging two different models provided here.

This study has limitations. The training centres very probably overlap with the cohort on which the IPSS-M was developed [3]; if so, the IPSS-M's cross-validated scores are partly in-sample and optimistic, and the external centre is its fairest test. External validation rests on one centre, scored on a public subset of its patients, and the platform reports no confidence interval for that score. The data contain no age, treatment or disease subtype, all of which affect survival. Monocytes and 60 genes could not be used because of differences between centres, and laboratory calibration differences cannot be checked. The karyotype parser is a set of hand-written rules rather than a validated nomenclature parser. The four-variable benchmark was reconstructed from the challenge description. Finally, the boosted model's results differ in the fourth decimal place between operating systems, although each machine reproduces its own results exactly.

Conclusion

For myeloid neoplasms, fitted survival models built on clinical, cytogenetic and molecular data, and validated on hospitals they had not seen, matched the published IPSS-M in cross-validation and exceeded it by 0.030 at one external centre, a difference not explained by case mix or missing data but not yet confirmed beyond that centre. Claims that a new model improves prognosis should be tested against the IPSS-M, with validation grouped by centre and paired comparisons, before they are accepted.

Supporting information

S1 Table. TRIPOD+AI checklist. Location in the manuscript of each item of the TRIPOD+AI reporting guideline.

S1 Appendix. Predictor definitions. Karyotype parsing rules, IPSS-R points and cytogenetic categories, IPSS-M coefficients and reference means, the shared gene panel and the pathway gene lists.

S1 Fig. Cox models on each validation fold. Concordance of each Cox model on each of the 15 centre-grouped folds, and the paired difference with the IPSS-R score.

S2 Fig. Tree models on each validation fold. Concordance of the random survival forest and the boosted Cox model on each fold, and the paired difference with the elastic-net.

S3 Fig. Time-varying effects. Scaled Schoenfeld residuals plus the estimated coefficient for NPM1 mutation and platelets, against the constant effect assumed by the Cox model.

S4 Fig. Time-dependent area under the curve. Cumulative/dynamic area under the curve at 1, 3 and 5 years for each model (mean of 15 folds).

S1 File. Full technical report. Every analysis stage, with all intermediate tables and figures.

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